Final Project Report

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Data:

CSV File link:
https://www.kaggle.com/datasets/nelgiriyewithana/most-streamed-spotify-songs-2024/data

- This had over 1000 + rows and the content in this CSV file included, artists, track, release date, spotify streams, youtube views, tiktok posts and many other platforms and the numbers they did on it.

Spotify API: https://developer.spotify.com/documentation/web-api/

- From the API I got popularity, track id, album name, track name and release date. It was used to get the top tracks from 2025 from the most streamed artists in 2024

Data Integration:

- As I stated previously, I used the Spotify API to collect the current top 5 tracks for the top 5 artists of 2024. Then to get the top 5 artists of 2024 I used the CSV that I found online to match them. They were matched using track_name and artists after cleaning and formatting them. Then I merged them into a dataframe with popularity, release dates and other columns.

Cleaning Challenges:

- There were a couple of challenges I faced when coding in the notebook. For instance the first error I faced was that there were many numeric fields with commas so I had to make sure I converted them into integers
- Then some of the release_date values were handled with errors so I used the feedback from the notebook and fixed it with “errors = ‘coerce’ ”
- Then some of the track names on Spotify did not match exactly with the CSV so I had to look up online some ways to fix this and found that .str.strip() can lowercase so that we can merge more accurately
- I needed to convert some datatype into BOOLEAN
- Then in the API it had a different format for release_date so I had to parse it all to the standard format

RDB Schema:

- For the actual tables the code will be written in a sql file and be submitted but the tables are artists, tracks, and StreamingMetrics
- Then for the relationships and diagram, I am going to be using an ERD Diagram to showcase the relationships between tables. To do this I am going to be using a website called AIAssist. I am enrolled in 480 this semester with professor Riazi so I don’t think

there will be any issues with me using it to generate me a diagram. I will provide the notation and diagram.

Notation:
[Entity]
Artists
--
artists_id @key
artists_name

[Entity]
Tracks
--
track_id @key
track_name
album_name
release_date
popularity
explicit
artist_id

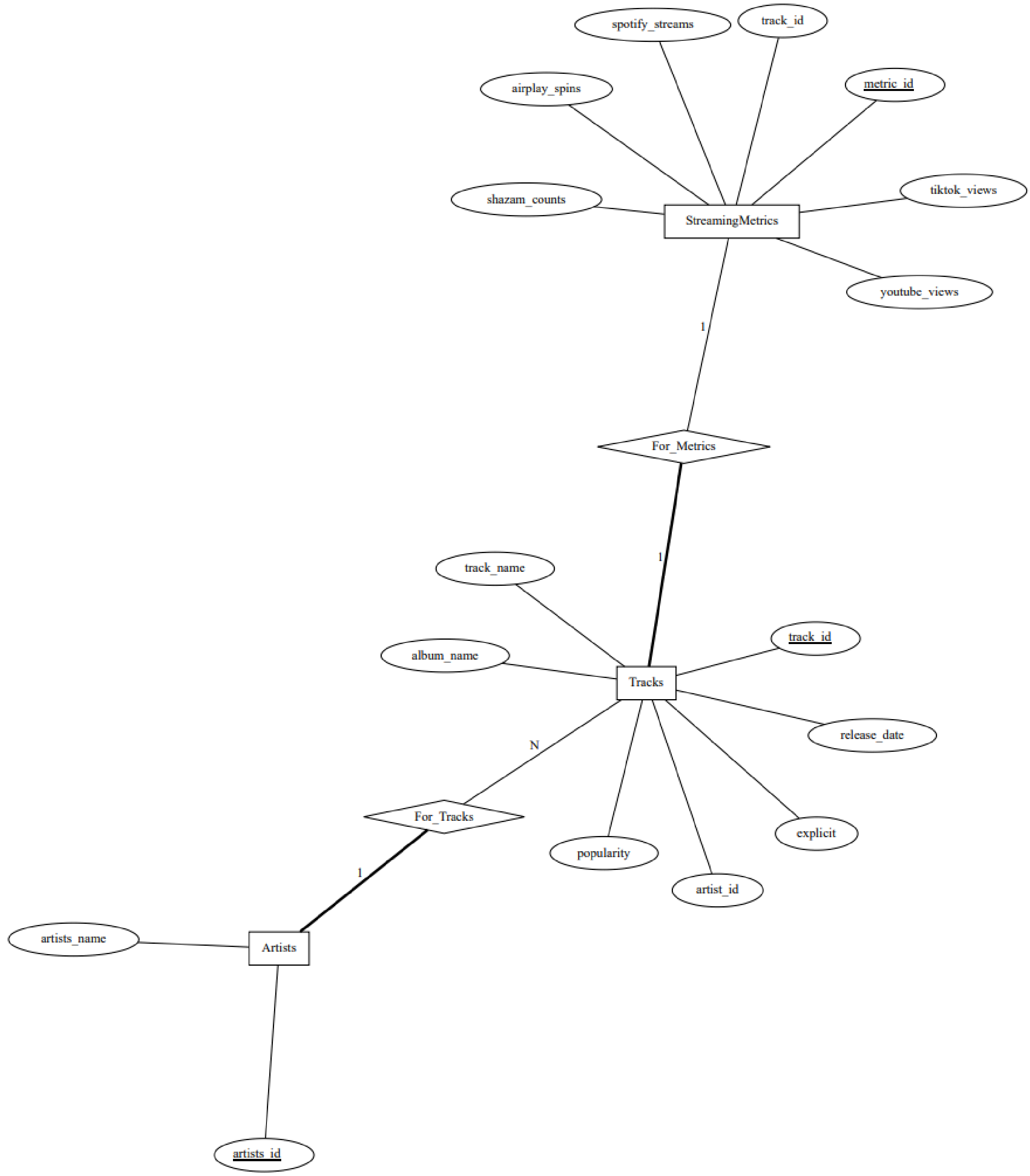
[Entity]
StreamingMetrics
--
metric_id @key
track_id
spotify_streams
youtube_views
tiktok_views
airplay_spins
shazam_counts

[Relationship]
For_Tracks
--
Artists (1) @total_participation

Tracks (N)

[Relationship]
For_Metrics
--
Tracks (1) @total_participation
StreamingMetrics (1)

ERD Diagram:



Queries and Plots:

- For queries I have create the required 5. One gets the total number of tracks per artist. Then another gets the Top 3 Streamed Tracks from 2024. Then another gets the most recent tracks released. Then one gets the average of popularity by artist in 2025. Lastly, this one measure the top 5 tracks on Shazam from the 2024 CSV.

- As for plots one is for the number of tracks per year, then another is for spotify streams, then lastly there is one for popularity per year.
- I was going to submit images of the queries and plots in the presentation slide so in my submission I will include it in there so that it is not in the report and it is repetitive.